# Explainable Machine Learning-Based Nowcasting of Fog and Low Visibility at Ankara Esenboğa Airport

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***Abstract*—Fog is the main meteorological cause of delays, diversions and cancellations at Ankara Esenboğa International Airport (LTAC), where more than half of all fog observations require CAT II/III approach procedures. This study presents the first machine-learning nowcasting system for low visibility at LTAC together with a benchmark of thirteen techniques from six families. Using 21.8 years (2005–2026) of quality-controlled METAR/SPECI reports, fog (prevailing visibility below 1000 m) and ILS-oriented runway visual range categories are predicted 1, 3 and 6 h ahead from 77 predictors that describe the current state, tendencies, boundary-layer memory and the solar cycle. Logistic regression, an analog (k-nearest-neighbour) forecast, random forest, XGBoost, LightGBM, CatBoost, an explainable boosting machine (EBM), a multilayer perceptron, GRU, LSTM, temporal convolutional and Transformer networks and a greedy ensemble were trained chronologically, calibrated and verified on four independent fog seasons (2022–2026) against persistence, a forecaster's rule of thumb, climatology and the official METAR TREND, with Holm-corrected day-block bootstrap tests. The ensemble ranks first overall; the best critical success index is 0.73, 0.47 and 0.34 at 1, 3 and 6 h, compared with 0.65, 0.32 and 0.13 for persistence, and at 1 h LightGBM exceeds the TREND by 0.106 (95% confidence interval 0.073–0.138). Tree ensembles and the glass-box EBM lead at 1 h and recurrent networks at 3–6 h, but the differences among the leading techniques are mostly not significant, whereas all of them clearly beat the reference forecasts and reach a relative economic value of 0.70–0.89 for a cost–loss ratio of 0.05. SHAP analysis reveals a physically consistent shift from the current humidity and visibility state at 1 h to boundary-layer moisture memory and the time to sunrise at 6 h. The calibrated models are deployed in an open-source web application that issues tiered warnings, explains every forecast and verifies itself in real time.**

***Index Terms*—Aviation meteorology, ensemble learning, explainable artificial intelligence, fog, gradient boosting, low-visibility procedures, nowcasting, recurrent neural networks, runway visual range.**

## Plain-Language Summary

FOG stops normal landings. At Ankara Esenboğa Airport we taught computers to forecast it from 21 years of the airport's half-hourly weather reports and compared thirteen machine-learning methods with the forecasts used today. One hour ahead, the best method (an ensemble of several models) knew in advance 83 of every 100 fogs, and 86 of every 100 of its warnings came true; simply assuming that the current weather persists caught 79 and was right 79 times, and the official short-range forecast did slightly worse. Six hours ahead the gap widens: the ensemble caught 54 of 100 fogs, persistence only 23. No single method won everywhere—tree-based methods were best for the next hour, networks with a memory for 3–6 h—but combining them gave the most reliable results. The methods learnt the physics of fog (moist air, calm wind, long nights) rather than arbitrary patterns, and a free public website runs them every 30 minutes and checks its own forecasts.

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## I. Introduction

Fog reduces the horizontal visibility near the ground below 1000 m and is one of the most disruptive weather phenomena for aviation [1], [2]. When visibility or runway visual range (RVR) falls below the minima of the available instrument landing system (ILS) category, an airport must activate low-visibility procedures, reduce its arrival rate or close, which propagates delays, diversions and cancellations through the network [3]. Ankara Esenboğa International Airport (LTAC), the main gateway of the Turkish capital, is particularly exposed: about 89% of its fogs occur between October and February, most of them reduce RVR below 350 m and roughly three quarters are radiation fogs [4]; for 2011–2020 about 800 fog hours were recorded and a single episode in December 2019 lasted almost three days [5].

Despite this operational relevance, the published work on Esenboğa is climatological and synoptic. Çamalan et al. [4] analysed 2000–2009 observations and explicitly recommended the development of fog prediction models, and Koyuncu et al. [5] combined a decadal analysis with a synoptic case study. To the best of our knowledge, no machine-learning (ML) forecast of fog or low visibility has been developed for LTAC. Internationally, statistical and ML nowcasts of fog and low-visibility-procedure states have shown clear skill over persistence at airports such as Vienna [6]–[8] and in many other locations [9], [10], but few studies compare data-driven forecasts with the official aerodrome forecasts under an identical information horizon, and few explain what the models have learnt in physical terms.

This study closes these gaps for LTAC. Its contributions are:

1) a 21.8-year quality-controlled METAR/SPECI data set (2005–2026) whose fog statistics reproduce the published climatology of the airport, released openly together with all code;

2) a benchmark of thirteen techniques from six families—linear, analog, tree ensembles (random forest, XGBoost, LightGBM, CatBoost), a glass-box explainable boosting

machine, neural networks (MLP, GRU, LSTM, TCN, Transformer) and a greedy ensemble—for fog and ILS-category forecasts at 1, 3 and 6 h, with probability calibration, a class-imbalance ablation, Holm-corrected day-block bootstrap tests, the relative economic value and the computational cost, against persistence, a forecaster's rule of thumb, climatology and the official METAR TREND (and the TAF in an operational window);

3) an ordinal formulation of the RVR/ILS-category forecast and a dedicated analysis of fog onsets and dissipations, the events that matter most for operations;

4) a SHAP-based explanation that connects the learnt relationships to fog physics and shows how the dominant predictors change with lead time; and

5) an open-source web-based early-warning application that runs the identical pipeline on live METARs, issues tiered warnings with explanations and verifies its own forecasts in real time.

The remainder of the paper is organised as follows. Section II reviews related work, Section III describes the airport and the data, Section IV the methodology, Section V the results, and Section VI discusses implications and limitations; Section VII concludes.

## II. Related Work

### A. Fog Physics and Esenboğa

Fog forms when the near-surface air becomes saturated through radiative cooling, advection of moist air or moistening by evaporation, and dissipates through solar heating, mixing or drying [1]. Field campaigns such as ParisFog showed that the life cycle of radiation fog is controlled by a delicate balance of radiative cooling, turbulence, aerosols and surface moisture [2], which explains why numerical weather prediction still struggles with fog [1], [11]. At Esenboğa, Çamalan et al. [4] found that radiation fog accounts for 75% of the fog days of 2000–2009 and that most fog observations fall into the CAT III range (RVR < 350 m), and Koyuncu et al. [5] related a prolonged December 2019 event to a persistent anticyclone and a strong low-level inversion.

### B. Statistical and Machine-Learning Nowcasting at Airports

Early data-driven approaches used artificial neural networks for fog at Canberra [12], for ceiling and visibility at US airports [13] and for probabilistic visibility forecasts [14]. Herman and Schumacher [11] post-processed reforecasts to improve fog and visibility guidance for aviation. For Vienna International Airport, Kneringer et al. [6] nowcast ordered low-visibility-procedure states with ordered logistic regression up to 2 h ahead, and Dietz et al. showed that tree-based models [7] and boosting [8] improve such forecasts for different planning horizons. Guijo-Rubio et al. [15] treated fog-induced low visibility as an ordinal classification problem, and Castillo-Botón et al. [16] compared regression and classification formulations of many ML methods. Deep learning approaches range from deep networks for airport visibility [17] and climatological visibility forecasting [18] to the three-dimensional convolutional FogNet [19]. Recent reviews [9], [10] confirm the potential of these methods but highlight inconsistent verification practices, rare comparisons with official forecasts and the limited transferability of models between sites [20]. Analog methods, which issue the outcome observed after the most similar past situations, have a long tradition in aviation forecasting [21] and form a strong probabilistic baseline [22]. In the wider tabular-learning literature, tree ensembles still often outperform deep networks [23], pre-trained tabular foundation models have recently become competitive on small data sets [24], and generalised additive models learnt by boosting promise accuracy without opacity [25]. Which of these families is best for fog, and at which lead time, has not been tested under one protocol.

### C. Explainable AI and Forecast Verification

Explainability is essential for the acceptance of ML forecasts by forecasters [26]. SHAP values [27] provide additive, locally accurate attributions that can be computed exactly for tree ensembles [28]. Aerodrome forecasts are verified with categorical scores derived from contingency tables [29], [30]; because TAFs and TRENDs contain change groups, they are best interpreted as ranges of forecast conditions [31]. The present study combines these strands—calibrated ML, ordinal ILS categories, explicit comparison with the official forecasts and SHAP explanations—in a single, operationally deployed system for an airport that had no data-driven fog forecast.

## III. Study Area and Data

### A. Esenboğa Airport and Its Fog Climate

Ankara Esenboğa International Airport (40.13° N, 32.99° E) lies on a plateau about 28 km north-east of the city centre at an elevation of 953 m. Its continental climate combines long winter nights, frequent anticyclonic stagnation, snow cover and moist boundary layers after frontal precipitation—conditions that favour radiative cooling and fog [4], [5]. Fig. 1 summarises the fog climate in the data set used here. Fog is almost exclusively a cold-season phenomenon: 94.6% of all half-hourly observations with a prevailing visibility below 1000 m fall between October and February, with frequencies of 4.3% in December and 3.7% in January and practically none from June to September. Within the day, fog is most frequent around sunrise (08 LT, 2.8%) and least frequent at 15 LT. The 819.5 fog hours we count for 2011–2020 agree closely with the approximately 800 h reported for that decade by Koyuncu et al. [5], and the share of fog observations with RVR ≤ 350 m (55.5%) confirms the finding of Çamalan et al. [4] that most fogs require CAT III operations. The 537 fog episodes are short—median 2 h—but the 90th percentile reaches 9 h and the longest lasted 35.5 h, so the operational problem is dominated by the timing of onsets and dissipations. The annual number of fog hours has decreased markedly: from about 130 h per year in 2005–2012 to about 50 h per year in 2020–2025.

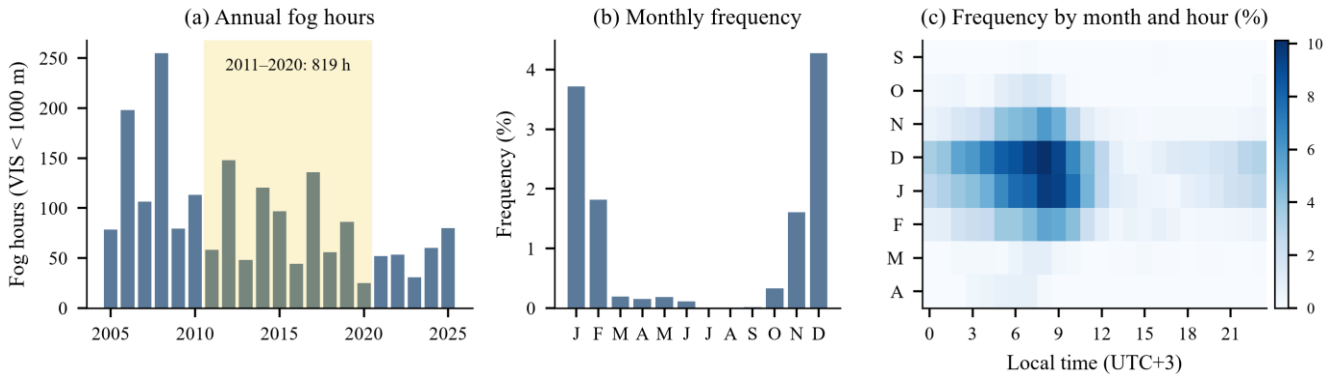



Fig. 1. Fog climate of LTAC, 2005–2026 (prevailing visibility < 1000 m). (a) Annual number of fog hours; the shaded decade 2011–2020 is the period analysed by Koyuncu et al. (b) Monthly frequency of fog in the half-hourly observations. (c) Frequency by month and local hour (UTC+3).

### *B. Observations and Official Forecasts*

The study uses the routine half-hourly METARs (issued at HH:20 and HH:50 UTC) and special SPECI reports of January 2005–October 2026. Original report texts were retrieved from the Iowa Environmental Mesonet (IEM) [32]. Older IEM holdings also contain reports reconstructed from the NCEI Integrated Surface Database; as these lack RVR and TREND groups and carry visibility in statute-mile steps, they were discarded and the resulting gaps were filled with original texts from the Ogimet archive [33]. Every METAR includes the forecaster's 2-h TREND forecast (NOSIG, BECMG or TEMPO groups), which serves as the official very-short-range forecast. Terminal aerodrome forecasts (TAFs; 24-h validity, issued four times a day) are available from the NOAA Aviation Weather Center data service [34] for the most recent 30 days only, because the historical TAF archive could not be accessed reliably; they are therefore evaluated in the operational period.

All reports were decoded with a purpose-built WMO FM 15/FM 51 decoder [35] that extracts prevailing and minimum visibility, the RVR of every runway, present and recent weather, cloud layers and vertical visibility, temperature, dew point, QNH and the TREND/TAF change groups; it also repairs frequent keying errors (e.g., "NO SIG", "Q1 015"). Quality control removed NIL and malformed reports, applied physical range checks and a dew-point consistency check, and removed isolated spikes in temperature, dew point and pressure. Table I summarises the resulting data set.

TABLE I
CHARACTERISTICS OF THE LTAC DATA SET

| Quantity | Value |
|---|---|
| Period (UTC) | 01/2005 – 10/2026 |
| METAR / SPECI reports (after QC) | 381 408 / 3 610 |
| Sources: IEM / Ogimet (gap filling) | 364 028 / 20 990 |
| Reports removed or corrected by QC | 109 / 47 |
| 30-min analysis slots (with observation) | 382 607 (99.7%) |
| Slots with VIS < 1000 m (rate) | 3 884 (1.02%) |
| Fog episodes; median / P90 duration | 537; 2.0 h / 9.0 h |
| RVR reported when VIS < 1500 m | 99.9% |
| Fog slots with RVR ≤ 350 m | 55.5% |

### *C. Analysis Grid and Predictands*

The reports were mapped onto a regular 30-min grid aligned with the routine observation times. The state at grid time $t$ is the latest report issued within $(t-29\,\text{min}, t]$, so SPECIs between routine times are honoured but no value is ever interpolated or carried forward; 99.7% of the 382 607 grid slots contain an observation.

Two predictands are defined at the valid time $t+h$ for lead times $h \in \{1, 3, 6\}$ h. The binary fog event is a prevailing visibility below 1000 m, the ICAO/WMO fog threshold [36]. The ILS category is an ordinal variable based on the minimum RVR reported over all runways: class 0 for RVR ≥ 550 m (CAT I or better), class 1 for 350 ≤ RVR < 550 m (CAT II) and class 2 for RVR < 350 m (CAT III). These boundaries follow the categories used for LTAC by Çamalan et al. [4] and the ICAO guidance on all-weather operations and RVR reporting [37], [38]; current CAT II minima allow RVR down to 300 m, so the 350-m boundary is slightly conservative. RVR is reported only when visibility or RVR is below 1500 m; slots without an RVR group and with visibility ≥ 1500 m are therefore class 0, and in the 0.06% of low-visibility slots with a missing RVR the prevailing visibility is used as a conservative proxy.

## IV. METHODOLOGY

Fig. 2 shows the data flow from the archives to the verified forecasts and to the operational application.

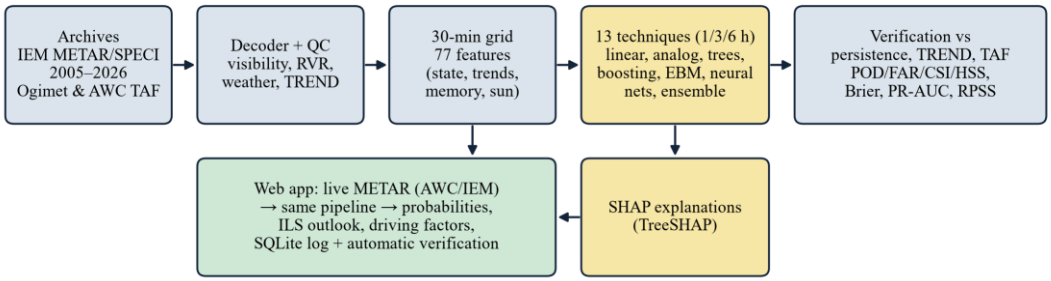



Fig. 2. Data flow of the study and of the operational early-warning application. The web application reuses the decoder, quality control, feature pipeline and calibrated models of the offline study, which guarantees identical offline and online forecasts.

### *A. Problem Formulation*

For every grid time $t$ with an observation, a model $f_h$ estimates the probability

$$p_h(t) = P(y(t+h) = 1 \mid \mathbf{x}(t)) \qquad (1)$$

where $\mathbf{x}(t)$ is a feature vector computed exclusively from reports issued at or before $t$. Warnings are issued when $p_h(t)$ exceeds a threshold $\tau_h$. The ILS category is forecast with the ordinal decomposition of Frank and Hall [39]: two binary models estimate $P_{550}=P(\text{RVR}<550\text{ m})$ and $P_{350}=P(\text{RVR}<350\text{ m})$, from which the class probabilities $1-P_{550}$, $P_{550}-P_{350}$ and $P_{350}$ follow, with $P_{350} \le P_{550}$ enforced.

### *B. Feature Engineering*

The 77 predictors are summarised by group in Table II. They describe (i) the current state—visibility, RVR, temperature $T$, dew point $T_d$, dew-point depression $T-T_d$, relative humidity from the Magnus formula [40], wind components, a calm flag, QNH, ceiling, cloud base and cover; (ii) present weather (fog, shallow or patchy fog, mist, haze, drizzle, rain, snow); (iii) 1-, 3- and 6-h tendencies, in particular the nocturnal cooling and the drying or moistening of the surface layer; (iv) memory terms such as the cooling since the 12-h temperature maximum, the fractions of fog, mist, precipitation and snow over the previous 6–72 h and the time since the last fog; and (v) calendar and astronomical terms, including the solar elevation and the time to sunrise at both the issue and the valid time, computed with the NOAA solar-position equations [41]. The valid-time astronomical terms are deterministic and therefore legitimately known in advance.

TABLE II
PREDICTOR GROUPS (77 FEATURES)

| Group | n | Content |
|---|---|---|
| Current state | 21 | VIS, RVR, T, Td, T−Td, RH, wind (u, v, calm, gust), QNH, ceiling, cloud base and cover, VV |
| Present weather | 11 | FG, MIFG/BCFG/PRFG, VCFG, FZFG, BR, HZ, DZ, RA, SN, recent precipitation |
| Tendencies | 19 | 1/3/6-h changes of T, Td, T−Td, log VIS; RH, wind, QNH (3/12 h), ceiling |
| Memory | 13 | cooling since 12-h max, min T−Td, max RH, fog/mist/precip./snow fractions, hours since fog, SPECI age |
| Calendar & sun | 13 | hour and day of year (sin/cos), solar elevation, night, time to sunrise / since sunset, and at valid time |

### *C. Models*

Thirteen techniques from six families compete under an identical protocol (same predictors, splits, calibration and threshold rule). *Linear:* logistic regression (LR) with standardised, median-imputed inputs [42]. *Analog:* a k-nearest-neighbour analog forecast in the spirit of the fuzzy-logic analog system of Hansen [21] and the analog ensemble [22]; the probability is the fraction of the $k$ most similar situations of the training archive (Euclidean distance between 13 standardised physical predictors, including the valid-time solar elevation) that were foggy $h$ hours later, with $k$ chosen on validation. *Tree ensembles:* random forests (RF) [43] and the gradient-boosting libraries XGBoost [44], LightGBM [45] and CatBoost [46]; the boosting libraries handle missing values natively and their hyper-parameters were tuned with 30 trials of the tree-structured Parzen estimator in Optuna [47], maximising the validation PR-AUC with early stopping. *Glass box:* an explainable boosting machine (EBM) [25], [48], a generalised additive model whose per-feature shape functions and ten strongest pairwise interactions are learnt by cyclic boosting, so that every forecast is an exact sum of readable contributions. *Neural networks:* a feed-forward multilayer perceptron (MLP) on the same 77 predictors, and four multi-task sequence networks that differ only in their encoder—long short-term memory (LSTM) [49], gated recurrent units (GRU) [50], a temporal convolutional network (TCN) with dilated causal convolutions [51] and a self-attention Transformer encoder [52]. Each sequence network reads the last 12 h (24 grid steps) of 22 standardised state variables plus a presence mask, concatenates its summary of the sequence with memory and valid-time features and predicts all nine tasks (three targets × three lead times) through a two-layer head; the masked, positively re-weighted binary cross-entropy was minimised with AdamW [53] and early stopping on the mean validation PR-AUC, and six architectures per encoder were compared on the validation seasons. *Ensemble:* a greedy forward ensemble selection with replacement [54] averages the calibrated probabilities of the members that most increase the validation PR-AUC. LR, RF, the MLP and the EBM used small validation grids. All models were implemented with scikit-learn [55], PyTorch [56] and the libraries cited; the selected settings are listed in the Appendix.

### D. Class Imbalance and Calibration

Fog occupies only 1.0% of the observations and RVR < 350 m only 0.6%. Four strategies were compared for LR and LightGBM [57]: no correction, class weighting, SMOTE oversampling [58] and random undersampling. Because re-weighting and resampling distort predicted probabilities, every model is recalibrated with either no mapping, Platt scaling [59] or isotonic regression [60]; the method is selected by leave-one-season-out cross-validation of the Brier score within the validation seasons [61] and then refitted on the whole validation set. The warning threshold $\tau_h$ maximises the critical success index (CSI) on the validation set; since $F_1$=2 CSI/(1+CSI), the same threshold also maximises $F_1$.

### E. Training Protocol and Leakage Control

The data were split chronologically and season-aligned (Table III): training 01/2005–06/2019, validation 07/2019–06/2022 (three fog seasons) and test 07/2022–06/2026 (four fog seasons); the period from July 2026 onwards is reserved for the operational evaluation. A random split would leak information because consecutive half-hourly samples of one fog episode are nearly identical [62]. All tuning, calibration and threshold decisions used the validation seasons only, and the test seasons were evaluated once. Automated tests verify that every feature at time $t$ is unchanged when all reports after $t$ are deleted and that a target never re-uses the observation at $t$.

TABLE III
CHRONOLOGICAL DATA PARTITION

| Set | Period | Slots | Fog | Rate |
|---|---|---|---|---|
| Training | 01/2005–06/2019 | 253 017 | 3 130 | 1.24% |
| Validation | 07/2019–06/2022 | 52 581 | 308 | 0.59% |
| Test | 07/2022–06/2026 | 70 073 | 446 | 0.64% |
| Operational | 07/2026–10/2026 | 5 749 | 0 | 0.00% |

### F. Reference Forecasts

Persistence forecasts the currently observed state for $t+h$. A forecaster's rule of thumb for radiation fog serves as a transparent physical reference: fog is forecast when the dew-point depression and the wind speed are below thresholds, optionally only for valid times at night and optionally combined with persistence; the thresholds are chosen on the validation seasons from a grid of 160 combinations by maximising CSI. Climatology is the conditional event frequency for the calendar month and UTC hour of the valid time in the training period and is the reference of the Brier skill score. The TREND appended to the METAR issued at $t$ is evaluated for $h \leq 2$ h, and the TAF forecast for $t+h$ is taken from the latest TAF issued at or before $t$, so that the forecaster's information horizon equals that of the models. TREND and TAF contain change groups and are therefore converted into a visibility range at the valid time [31]: the prevailing reading uses the base forecast, FM groups and completed BECMG changes, and the worst-case reading additionally takes the lower end of BECMG transitions and of TEMPO and PROB30/40 groups. A fog forecast is issued when the reading is below 1000 m.

### G. Verification

With $a$ hits, $b$ false alarms, $c$ misses and $d$ correct negatives, the categorical scores are [29], [30]

$$\mathrm{POD} = \frac{a}{a+c}, \mathrm{FAR} = \frac{b}{a+b}, \mathrm{CSI} = \frac{a}{a+b+c} \tag{2}$$

$$\mathrm{HSS} = \frac{2(ad-bc)}{(a+c)(c+d)+(a+b)(b+d)} \tag{3}$$

where the Heidke skill score (HSS) measures accuracy relative to random chance [63], [64]. Probabilistic forecasts are assessed with the area under the precision–recall curve (PR-AUC), which is informative for rare events [65], [66], the area under the ROC curve, and the Brier score and its skill score [67], [68]

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(p_i - o_i)^2, \mathrm{BSS} = 1 - \frac{\mathrm{BS}}{\mathrm{BS_{clim}}} \tag{4}$$

For the ILS categories the ranked probability score [69]

$$\mathrm{RPS} = \frac{1}{K-1}\sum_{k=1}^{K}(F_k - O_k)^2 \tag{5}$$

with cumulative forecast and observed distributions $F_k$ and $O_k$ ($K$=3) is averaged over the test set and expressed as a skill score (RPSS) against the month–hour class climatology. Because CSI and HSS degenerate as events become rarer, the symmetric extremal dependence index (SEDI) [70], computed from the hit rate $H$ (POD) and the false-alarm rate $F$, complements them:

$$\mathrm{SEDI} = \frac{\ln F - \ln H - \ln(1-F) + \ln(1-H)}{\ln F + \ln H + \ln(1-F) + \ln(1-H)} \tag{6}$$

The practical benefit of a forecast is expressed by its relative economic value $V$ in the static cost–loss model [71]: a user who pays a cost $C$ to protect against a loss $L$ acts when the calibrated probability exceeds $\alpha=C/L$, and $V=(E_{\mathrm{clim}}-E_{\mathrm{fc}})/(E_{\mathrm{clim}}-E_{\mathrm{perf}})$ compares the resulting mean expense with that of the better of always or never protecting ($V$=0) and with a perfect forecast ($V$=1). Because half-hourly samples are strongly autocorrelated, 95% confidence intervals are obtained with a block bootstrap that resamples whole days [72], [73], and differences between forecasts are tested with paired resamples of the same days (2000 resamples for the categorical scores, 300 for PR-AUC and BSS). For the benchmark, every technique is compared with the best one of its lead time on 1000 paired day resamples, and the two-sided p-values are adjusted for the twelve comparisons per lead time with the Holm procedure [74].

### H. Explainability

The LightGBM models are explained with SHAP values [27], computed exactly for trees with TreeSHAP [28]. For a forecast with log-odds $g(\mathbf{x})$ the contributions $\varphi_j$ satisfy local accuracy,

$$g(\mathbf{x}) = \phi_0 + \sum_{j=1}^{M}\phi_j \tag{7}$$

where $\varphi_0$ is the expected model output and $\varphi_j$ is the Shapley value of feature $j$ [75]. Global importance is summarised by the mean absolute SHAP value on a test sample containing all fog cases and 6000 random non-fog cases, and dependence plots relate the contributions to the physical drivers of fog [26]. Three of the thirteen techniques are explainable by construction and act as transparent cross-checks: the coefficients of LR, the shape functions of the EBM and, for

the analog forecast, the list of past situations judged most similar.

### *I. Early-Warning Application*

The operational system is a FastAPI web application (Fig. 17). Every ten minutes it downloads the METARs of the last 7 days from the NOAA Aviation Weather Center (with IEM as fallback), runs exactly the same decoding, quality-control, gridding and feature code as the offline study, and publishes calibrated fog probabilities for 1–6 h, an ILS-category outlook for 1, 3 and 6 h, an alert level and the leading SHAP contributions. Because TreeSHAP values are additive, the 77 contributions are also summed into ten plain-language factor groups (e.g., humidity, wind, time of day) so that non-specialist users can read the explanation. The alert is red when fog is observed or the 1–2-h probability exceeds its threshold, orange when only the 3–6-h forecasts exceed their thresholds, and yellow (watch) when any probability exceeds half of its threshold. Every forecast is stored and automatically verified once its valid time has passed, which provides a genuinely real-time verification record. A serverless public instance runs the same cycle every 30 min on GitHub Actions and publishes the dashboard at https://atasakik.github.io/esenboga-fog-nowcasting/. The runs start at HH:10 and HH:40 UTC, because the data service receives the routine reports of HH:20 and HH:50 8–17 min after their observation time; an external scheduler triggers them on time and GitHub's own, often late, scheduler serves as a fallback. Because runs can still be delayed or skipped, each run also computes the forecasts for the issue times skipped during roughly the previous three days from the reports observed up to those times (including any later corrections), each with its full 96-h feature history; these catch-up forecasts are verified like the others but reported separately from the real-time record.

## V. Results

### *A. Overall Skill*

Table IV lists the full scores of the best technique of each family and of the reference forecasts on the four test seasons. At 1 h the ensemble and LightGBM attain the best CSI (0.729) and HSS (0.842), and the ensemble the best PR-AUC (0.899) and Brier skill score (0.742), against a CSI of 0.652 for persistence. The margin grows with lead time: at 3 h the best CSI is 0.472 (LSTM) against 0.321 for persistence, and at 6 h 0.341 (ensemble) against 0.129. Persistence even becomes worse than climatology at 3 and 6 h (negative BSS), whereas all ML techniques keep a positive BSS at 6 h (0.13 for the analog forecast, 0.25–0.32 for the others). The ROC areas are close to 1 for every technique (0.97–0.99) because of the many easy non-fog cases, so the PR curves in Fig. 3 are the more informative view: at 1 h the leading techniques are nearly indistinguishable, while at 3 and 6 h the recurrent networks dominate a large part of the curve.

TABLE IV
VERIFICATION OF THE FOG (VIS < 1000 M) FORECASTS ON THE TEST SEASONS 2022/23–2025/26

| Lead | Forecast | PR-AUC | ROC-AUC | BS | BSS | POD | FAR | CSI | HSS |
|---|---|---|---|---|---|---|---|---|---|
| 1 h | Ensemble | **0.899** | **0.998** | **0.0016** | **0.742** | 0.825 | 0.138 | **0.729** | **0.842** |
| | LightGBM | 0.891 | **0.998** | **0.0016** | 0.739 | 0.825 | 0.138 | **0.729** | **0.842** |
| | LSTM | 0.885 | **0.998** | 0.0017 | 0.725 | 0.789 | 0.135 | 0.703 | 0.824 |
| | EBM | 0.889 | **0.998** | **0.0016** | 0.733 | 0.821 | 0.135 | 0.728 | 0.841 |
| | Logistic regression | 0.885 | 0.997 | 0.0017 | 0.722 | 0.767 | **0.128** | 0.690 | 0.815 |
| | Analog (k-NN) | 0.840 | 0.989 | 0.0019 | 0.693 | 0.814 | 0.207 | 0.671 | 0.802 |
| | Persistence | – | – | 0.0027 | 0.563 | 0.789 | 0.211 | 0.652 | 0.788 |
| | Forecaster's rule | – | – | 0.0089 | -0.444 | **0.852** | 0.594 | 0.379 | 0.546 |
| | TREND (prevailing) | – | – | 0.0029 | 0.529 | 0.753 | 0.217 | 0.623 | 0.767 |
| | TREND (worst case) | – | – | 0.0030 | 0.512 | 0.774 | 0.240 | 0.622 | 0.765 |
| 3 h | Ensemble | **0.678** | **0.995** | **0.0033** | **0.463** | 0.547 | **0.263** | 0.458 | 0.626 |
| | LightGBM | 0.657 | 0.994 | 0.0034 | 0.455 | 0.585 | 0.302 | 0.467 | 0.634 |
| | LSTM | 0.673 | 0.994 | **0.0033** | **0.463** | **0.612** | 0.328 | **0.472** | **0.639** |
| | EBM | 0.639 | 0.992 | 0.0034 | 0.439 | 0.570 | 0.345 | 0.438 | 0.607 |
| | Logistic regression | 0.637 | 0.993 | 0.0035 | 0.425 | 0.529 | 0.328 | 0.421 | 0.590 |
| | Analog (k-NN) | 0.558 | 0.982 | 0.0041 | 0.334 | 0.502 | 0.439 | 0.361 | 0.527 |
| | Persistence | – | – | 0.0065 | -0.063 | 0.487 | 0.513 | 0.321 | 0.483 |
| | Forecaster's rule | – | – | 0.0101 | -0.644 | 0.599 | 0.665 | 0.274 | 0.425 |
| 6 h | Ensemble | **0.516** | **0.990** | **0.0042** | 0.313 | **0.538** | 0.517 | **0.341** | **0.506** |
| | LightGBM | 0.457 | 0.988 | 0.0045 | 0.270 | 0.522 | 0.592 | 0.297 | 0.454 |
| | LSTM | 0.509 | 0.988 | **0.0042** | **0.314** | 0.475 | **0.456** | 0.340 | 0.504 |
| | EBM | 0.438 | 0.980 | 0.0046 | 0.252 | 0.392 | 0.524 | 0.274 | 0.427 |
| | Logistic regression | 0.453 | 0.987 | 0.0045 | 0.264 | 0.496 | 0.584 | 0.292 | 0.449 |
| | Analog (k-NN) | 0.264 | 0.968 | 0.0053 | 0.131 | 0.327 | 0.720 | 0.178 | 0.297 |
| | Persistence | – | – | 0.0098 | -0.598 | 0.229 | 0.771 | 0.129 | 0.224 |
| | Forecaster's rule | – | – | 0.0115 | -0.876 | 0.307 | 0.785 | 0.145 | 0.248 |

BS: Brier score; BSS relative to the month–hour climatology of the training period. Categorical scores use the validation-optimal (maximum-CSI) threshold. Best value per lead time in bold. TREND is only defined up to 2 h.

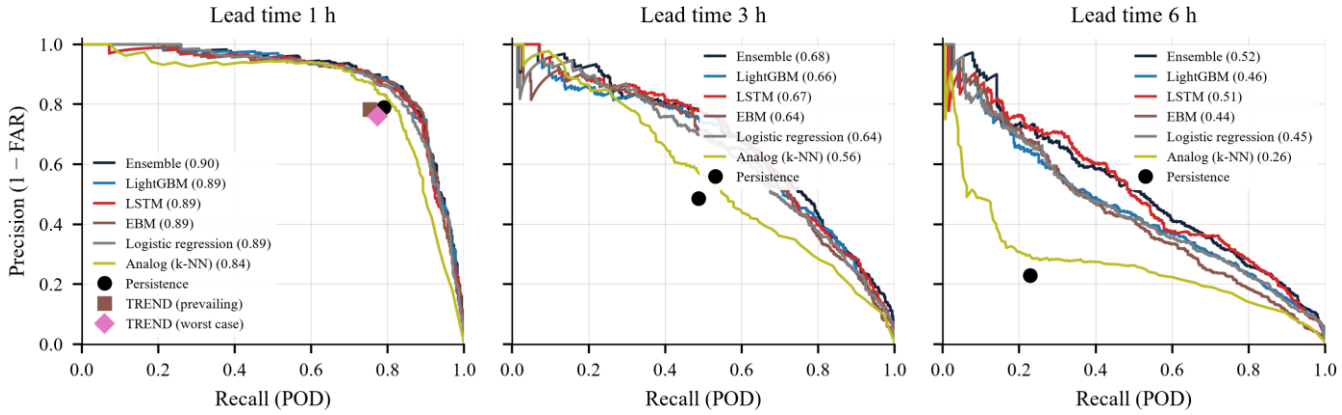



Fig. 3. Precision–recall curves of the best technique of each family for fog on the test seasons (PR-AUC in brackets). Markers show the deterministic reference forecasts.

The day-block bootstrap (Table VII) confirms that the improvements are robust: the paired CSI and HSS differences between the ML techniques and persistence are significantly positive at all three lead times—for the ensemble, for example, ΔCSI = 0.077 [0.054, 0.103] at 1 h, 0.136 [0.099, 0.173] at 3 h and 0.213 [0.143, 0.282] at 6 h. The skill decays smoothly with lead time (Fig. 7), and the relative CSI gain of LightGBM over persistence increases from 12% at 1 h to 45% at 3 h and 130% at 6 h. The improvement is not limited to a favourable year: all thirteen techniques beat persistence in each of the four test seasons and at every lead time, with a single exception—the analog forecast at 1 h in 2025/26 (Table VIII).

### *B. Benchmark of the Thirteen Techniques*

Table V and Fig. 4 compare all techniques. No family dominates at every lead time. At 1 h the ensemble, the tree ensembles and the glass-box EBM lead: the PR-AUC of the ensemble is significantly higher than that of every neural network, CatBoost, LR and the analog forecast (Holm-adjusted $p < 0.05$), but not than that of random forest, XGBoost, LightGBM and the EBM, and the EBM practically matches the best CSI (0.728). At 3 and 6 h the recurrent networks reach the highest PR-AUC (GRU: 0.685 and 0.517) and the LSTM the highest CSI at 3 h (0.472), but their advantage over the tree ensembles, LR, the EBM and the ensemble lies within the sampling uncertainty of four seasons; only the analog forecast is significantly worse at every lead time, joined by the MLP and the Transformer at 3 h. Averaged over the five scores and three lead times the ensemble ranks first (mean rank 2.3), followed by LightGBM (3.9), LSTM (4.5) and random forest (4.8); the simplest learner, LR, ranks eleventh, yet still far ahead of every reference forecast. The greedy selection picked different members at each lead time—XGBoost, LSTM, CatBoost and GRU at 1 h, seven members from five families at 3 h, and GRU, XGBoost, Transformer and CatBoost at 6 h—so its consistency comes from diversity across families rather than from a single strong model.

TABLE V
BENCHMARK OF THE THIRTEEN TECHNIQUES ON THE TEST SEASONS 2022/23–2025/26 (FOG, VIS < 1000 M)

| Technique | Family | PR-AUC 1 h | PR-AUC 3 h | PR-AUC 6 h | CSI 1 h | CSI 3 h | CSI 6 h | Mean rank |
|---|---|---|---|---|---|---|---|---|
| Ensemble | ensemble | **0.899** | 0.678 | 0.516 | **0.729** | 0.458 | **0.341** | 2.3 |
| LightGBM | trees | 0.891 | 0.657 | 0.457 | **0.729** | 0.467 | 0.297 | 3.9 |
| LSTM | neural | 0.885† | 0.673 | 0.509 | 0.703 | **0.472** | 0.340 | 4.5 |
| Random forest | trees | 0.889 | 0.665 | 0.470 | 0.715 | 0.470 | 0.309 | 4.8 |
| GRU | neural | 0.882† | **0.685** | **0.517** | 0.716 | 0.457 | 0.290 | 6.0 |
| XGBoost | trees | 0.892 | 0.657 | 0.471 | 0.713 | 0.453 | 0.308 | 6.2 |
| TCN | neural | 0.884† | 0.643 | 0.486 | 0.715 | 0.457 | 0.329 | 6.3 |
| CatBoost | trees | 0.886† | 0.656 | 0.455 | 0.702† | 0.459 | 0.300 | 6.9 |
| EBM | glass box | 0.889 | 0.639 | 0.438 | 0.728 | 0.438 | 0.274 | 7.5 |
| Transformer | neural | 0.880† | 0.626† | 0.447 | 0.717 | 0.415 | 0.288 | 9.4 |
| Logistic regression | linear | 0.885† | 0.637 | 0.453 | 0.690† | 0.421 | 0.292 | 9.7 |
| MLP | neural | 0.882† | 0.604† | 0.439 | 0.703 | 0.437 | 0.250 | 10.7 |
| Analog (k-NN) | analog | 0.840† | 0.558† | 0.264† | 0.671† | 0.361† | 0.178† | 12.6 |
| TREND (prevailing) | reference | – | – | – | 0.623 | – | – | – |
| Forecaster's rule | reference | – | – | – | 0.379 | 0.274 | 0.145 | – |
| Persistence | reference | – | – | – | 0.652 | 0.321 | 0.129 | – |

Rows sorted by the mean rank over PR-AUC, BSS, CSI, HSS and SEDI at 1, 3 and 6 h. Best technique per column in bold; † significantly worse than the best technique of that lead time (paired day-block bootstrap, 1000 resamples, Holm-adjusted $p < 0.05$). Reference forecasts are deterministic, so PR-AUC is not defined; TREND only up to 2 h.

| | PR-AUC | | | CSI | | | SEDI | | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 h | 3 h | 6 h | 1 h | 3 h | 6 h | 1 h | 3 h | 6 h |
| Ensemble | **0.899** | 0.678 | 0.516 | **0.729** | 0.458 | **0.341** | **0.957** | 0.850 | **0.822** |
| LightGBM | 0.891 | 0.657 | 0.457 | **0.729** | 0.467 | 0.297 | **0.957** | 0.863 | 0.805 |
| LSTM | 0.885 | 0.673 | 0.509 | 0.703 | **0.472** | 0.340 | 0.947 | 0.872 | 0.797 |
| Random forest | 0.889 | 0.665 | 0.470 | 0.715 | 0.470 | 0.309 | 0.949 | 0.860 | 0.790 |
| GRU | 0.882 | **0.685** | **0.517** | 0.716 | 0.457 | 0.290 | 0.953 | 0.849 | 0.741 |
| XGBoost | 0.892 | 0.657 | 0.471 | 0.713 | 0.453 | 0.308 | 0.948 | 0.845 | 0.787 |
| TCN | 0.884 | 0.643 | 0.486 | 0.715 | 0.457 | 0.329 | 0.948 | **0.880** | 0.788 |
| CatBoost | 0.886 | 0.656 | 0.455 | 0.702 | 0.459 | 0.300 | 0.952 | 0.859 | 0.779 |
| EBM | 0.889 | 0.639 | 0.438 | 0.728 | 0.438 | 0.274 | 0.956 | 0.853 | 0.744 |
| Transformer | 0.880 | 0.626 | 0.447 | 0.717 | 0.415 | 0.288 | 0.952 | 0.851 | 0.774 |
| Logistic regression | 0.885 | 0.637 | 0.453 | 0.690 | 0.421 | 0.292 | 0.941 | 0.836 | 0.792 |
| MLP | 0.882 | 0.604 | 0.439 | 0.703 | 0.437 | 0.250 | 0.943 | 0.844 | 0.709 |
| Analog (k-NN) | 0.840 | 0.558 | 0.264 | 0.671 | 0.361 | 0.178 | 0.951 | 0.813 | 0.667 |
| TREND | – | – | – | 0.623 | – | – | 0.931 | – | – |
| Forecaster's rule | – | – | – | 0.379 | 0.274 | 0.145 | 0.951 | 0.835 | 0.634 |
| Persistence | – | – | – | 0.652 | 0.321 | 0.129 | 0.943 | 0.796 | 0.580 |

Fig. 4. Test-season scores of the thirteen techniques and the reference forecasts for fog at 1, 3 and 6 h. Colours show the position within each column (darker is better); the best technique per column is in bold. Techniques are sorted by their mean rank.

The forecaster's rule of thumb selected on the validation seasons—fog when the dew-point depression is at most 0.5 °C and the wind at most 3 kt at a night-time valid time, or when fog is observed now—detects many fogs (POD 0.85 at 1 h) but at the price of a false-alarm ratio of 0.59, so its CSI (0.38, 0.27 and 0.15) stays below that of persistence at 1 and 3 h. Its high SEDI at 1 h (0.95) shows that SEDI rewards detection largely irrespective of false alarms and must be read together with CSI.

The relative economic value (Fig. 5) translates these scores into decisions. For users with a low cost–loss ratio, for whom protective action is cheap compared with the loss caused by an unforeseen fog, the ensemble achieves $V = 0.89$, 0.77 and 0.70 at $\alpha = 0.05$ for 1, 3 and 6 h, against 0.78, 0.46 and 0.19 for persistence and 0.74 for the TREND at 1 h. At $\alpha = 0.2$ the ensemble retains 0.81, 0.55 and 0.39, whereas persistence falls to 0.36 and 0.04 at 3 and 6 h and the rule of thumb loses its value because of its false alarms. Because every user can choose the probability threshold that matches their own cost–loss ratio, the calibrated techniques stay valuable over a much wider range of cost–loss ratios than the deterministic reference forecasts.

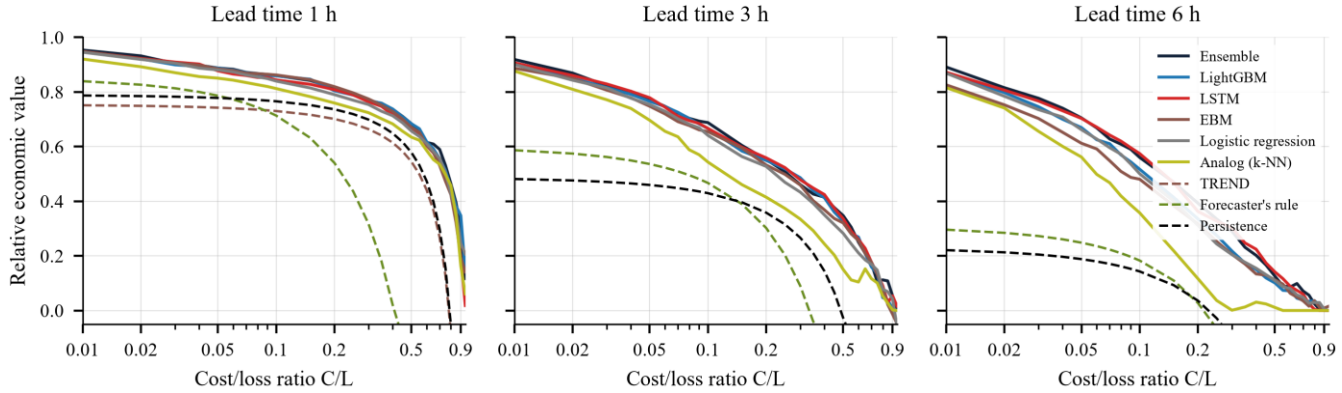



Fig. 5. Relative economic value of the best technique of each family and of the reference forecasts as a function of the user's cost–loss ratio (the probabilistic forecasts are used with the threshold $p \geq \alpha$).

The techniques differ far more in cost and transparency than in skill (Table VI, Fig. 6). LightGBM fits in 3 s and LR in 5 s, the other boosting libraries and the MLP in 20–35 s, the multi-task sequence networks in 26–95 s for all nine tasks together and the EBM in 275 s; all techniques except the analog forecast, which searches the whole training archive for every forecast, predict 10 000 cases in less than 0.15 s. LR and the EBM explain themselves exactly, the analog forecast through its past cases and the tree ensembles through exact TreeSHAP values, whereas the neural networks and the ensemble need approximate post hoc methods. Since the EBM loses only 0.001 in CSI to the best technique at 1 h, transparency is almost free at the shortest lead time; at 3–6 h the opaque recurrent networks and the ensemble add 0.05–0.08 in PR-AUC.

TABLE VI
COMPUTATIONAL COST AND EXPLAINABILITY

| Technique | Fit (s) | Predict (ms/10k) | Size (MB) | Explanation |
|---|---|---|---|---|
| Logistic regression | 5 | 21 | 0.0 | coefficients (built in) |
| Analog (k-NN) | 0 | 3822 | 2.3 | the past analog cases (built in) |
| Random forest | 33 | 148 | 7.7 | SHAP (post hoc) |
| XGBoost | 26 | 22 | 0.3 | SHAP (post hoc) |
| LightGBM | 3 | 15 | 0.2 | SHAP (post hoc) |
| CatBoost | 21 | 13 | 0.3 | SHAP (post hoc) |
| EBM | 275 | 9 | 0.8 | per-feature shape functions (built in) |
| MLP | 21 | 17 | 0.8 | post hoc only |
| GRU | 47 | – | 0.1 | post hoc only |
| LSTM | 26 | – | 0.3 | post hoc only |
| TCN | 95 | – | 0.2 | post hoc only |
| Transformer | 43 | – | 0.2 | post hoc only |

Fit: one fit of the selected configuration (fog, 3 h; the sequence networks fit all nine tasks at once) on a laptop with an 8-GB GPU; the hyper-parameter search is not included. Predict: 10 000 test rows on the same machine.

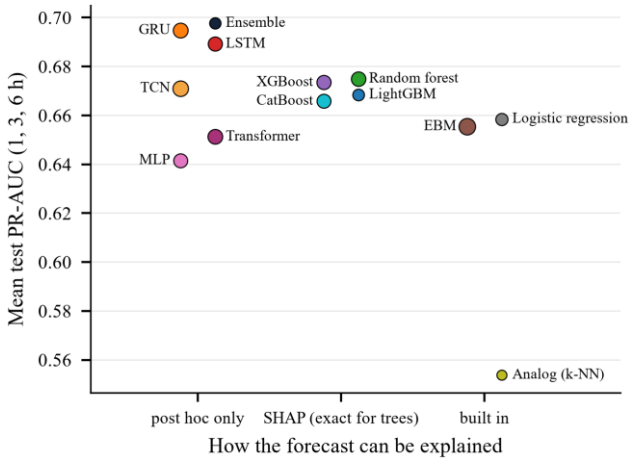



Fig. 6. Mean test PR-AUC over 1, 3 and 6 h against the way each technique can be explained; the marker size grows with the fit time.

TABLE VII
95% DAY-BLOCK BOOTSTRAP CONFIDENCE INTERVALS (TEST SEASONS)

| Lead | Forecast | CSI | HSS | PR-AUC |
|---|---|---|---|---|
| 1 h | Ensemble | 0.73* [0.67, 0.78] | 0.84 [0.80, 0.88] | 0.90 [0.84, 0.93] |
| | LightGBM | 0.73* [0.66, 0.78] | 0.84 [0.80, 0.88] | 0.89 [0.84, 0.93] |
| | Persistence | 0.65 [0.59, 0.71] | 0.79 [0.74, 0.83] | – |
| | Forecaster's rule | 0.38 [0.30, 0.45] | 0.55 [0.45, 0.62] | – |
| 3 h | Ensemble | 0.46* [0.36, 0.54] | 0.63 [0.53, 0.70] | 0.68 [0.55, 0.78] |
| | LightGBM | 0.47* [0.38, 0.54] | 0.63 [0.54, 0.70] | 0.66 [0.53, 0.75] |
| | Persistence | 0.32 [0.24, 0.39] | 0.48 [0.38, 0.56] | – |
| | Forecaster's rule | 0.27 [0.20, 0.34] | 0.43 [0.34, 0.50] | – |
| 6 h | Ensemble | 0.34* [0.26, 0.42] | 0.51 [0.41, 0.59] | 0.52 [0.38, 0.62] |
| | LightGBM | 0.30* [0.22, 0.38] | 0.45 [0.36, 0.54] | 0.46 [0.34, 0.57] |
| | Persistence | 0.13 [0.07, 0.19] | 0.22 [0.12, 0.32] | – |
| | Forecaster's rule | 0.14 [0.09, 0.20] | 0.25 [0.16, 0.33] | – |

*CSI significantly higher than persistence (paired bootstrap, 2000 resamples of days)*

TABLE VIII
SEASON-BY-SEASON SKILL AT 3 H (TEST)

| Season | Fog slots | CSI Ensemble | CSI LightGBM | CSI rule | CSI persist. | PR-AUC Ensemble |
|---|---|---|---|---|---|---|
| 2022/23 | 83 | 0.417 | 0.414 | 0.265 | 0.317 | 0.643 |
| 2023/24 | 94 | 0.407 | 0.432 | 0.199 | 0.253 | 0.591 |
| 2024/25 | 190 | 0.495 | 0.502 | 0.368 | 0.397 | 0.749 |
| 2025/26 | 79 | 0.475 | 0.486 | 0.205 | 0.244 | 0.711 |

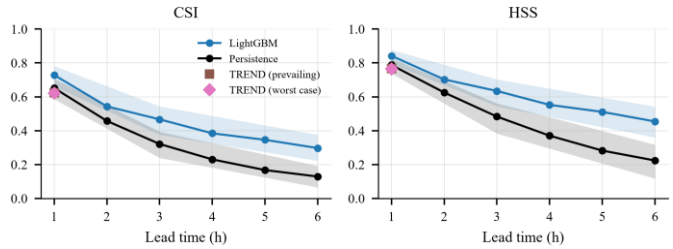



Fig. 7. CSI and HSS of LightGBM and persistence as a function of lead time on the test seasons, with the METAR TREND forecast at 1 h. Shading: 95% day-block bootstrap intervals computed at 1, 3 and 6 h (interpolated in between).

### C. Comparison with the Official TREND and TAF

The TREND forecast is available for every test METAR. On identical cases (Table IX), the prevailing reading of the TREND reaches CSI 0.623 and HSS 0.767 at 1 h and the worst-case reading 0.622 and 0.765, slightly below persistence (0.652 and 0.788). In other words, the official 2-h forecast rarely adds information beyond the current observation: TREND groups most often announce improvements, while new fog formation is seldom anticipated. LightGBM, the model deployed in the web application, improves on the TREND by $\Delta$CSI = 0.106 [0.073, 0.138] and $\Delta$HSS = 0.076 [0.052, 0.103], both significant; the ensemble reaches the same CSI on these cases. In the 30-day TAF window available at the time of writing (29 September–30 October 2026; about 1 420 forecasts per lead time), no fog occurred; neither the TAF—under either reading—nor LightGBM issued a fog forecast, so both were free of false alarms, while the worst-case TAF reading announced mist below 5 km in about 8% of the cases. A fog-season TAF comparison is therefore left to the live verification.

TABLE IX
ML VERSUS THE METAR TREND FORECAST ON IDENTICAL TEST CASES

| Lead | Forecast | POD | FAR | CSI | HSS |
|---|---|---|---|---|---|
| 1 h | Ensemble | **0.825** | **0.138** | **0.729** | **0.842** |
| | LightGBM | **0.825** | **0.138** | **0.729** | **0.842** |
| | Persistence | 0.789 | 0.211 | 0.652 | 0.788 |
| | TREND (prevailing) | 0.753 | 0.217 | 0.623 | 0.767 |
| | TREND (worst case) | 0.774 | 0.240 | 0.622 | 0.765 |

70 037 forecasts with a decodable TREND (446 fog cases at 1 h).

### D. Fog Onsets and Dissipations

Onsets and dissipations are where a nowcast creates operational value, because persistence cannot predict either. Fig. 8 shows the fraction of the 48 test-season onsets (fog after at least 2 h without fog) and 44 dissipations that the best technique of each family forecast correctly. Dissipations are captured well: at 1 h the EBM forecasts 41% of them correctly and LightGBM 34% (persistence: 0%), and at 6 h the LSTM and the EBM reach 77% (persistence: 68%). Onsets remain the hardest problem: at the CSI-optimal thresholds these techniques capture 2–23% of the onsets (the ensemble 23% at 6 h), compared with 0–4% for persistence. The tiered alert design mitigates this limitation: at the watch level ($\tau_h$/2) LightGBM flags 25%, 21% and 38% of the onsets 1, 3 and 6 h ahead, while the CSI decreases only from 0.729 to 0.692, from 0.467 to 0.435 and from 0.297 to 0.271.

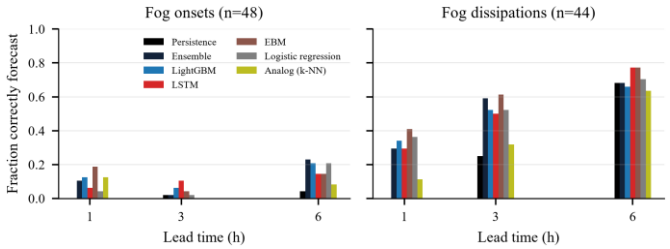



Fig. 8. Fraction of fog onsets (left) and dissipations (right) in the test seasons that were correctly forecast at the CSI-optimal thresholds.

### *E. ILS-Category Forecasts*

Table X and Fig. 9 summarise the ordinal RVR forecasts. All thirteen techniques achieve a positive ranked probability skill score at every lead time—the ensemble 0.642, 0.433 and 0.284 at 1, 3 and 6 h—whereas persistence falls from 0.427 to −0.166 and −0.695. For CAT III conditions (RVR < 350 m) LightGBM reaches CSI 0.578 at 1 h and the ensemble 0.381 at 3 h (persistence: 0.521 and 0.225). The confusion matrices show that CAT III is recognised well (76% of the cases at 1 h), whereas the intermediate CAT II class is the most difficult, being frequently confused with its neighbours; this is expected for a narrow transitional band of RVR.

TABLE X
ILS-CATEGORY (RVR) FORECASTS ON THE TEST SEASONS

| Lead | Forecast | RPSS | Macro-F1 | $HSS_3$ | $CSI_{\geq II}$ | $CSI_{III}$ |
|---|---|---|---|---|---|---|
| 1 h | Ensemble | **0.642** | 0.669 | 0.696 | 0.658 | 0.572 |
| | LightGBM | 0.636 | **0.701** | **0.706** | 0.644 | **0.578** |
| | LSTM | 0.620 | 0.629 | 0.680 | **0.660** | 0.532 |
| | LR | 0.596 | 0.602 | 0.656 | 0.632 | 0.514 |
| | Analog (k-NN) | 0.571 | 0.608 | 0.641 | 0.604 | 0.507 |
| | Persistence | 0.427 | 0.673 | 0.657 | 0.599 | 0.521 |
| 3 h | Ensemble | **0.433** | 0.540 | **0.529** | **0.462** | **0.381** |
| | LightGBM | 0.393 | **0.568** | 0.501 | 0.427 | 0.341 |
| | LSTM | 0.393 | 0.552 | 0.494 | 0.446 | 0.319 |
| | LR | 0.345 | 0.525 | 0.468 | 0.407 | 0.311 |
| | Analog (k-NN) | 0.280 | 0.484 | 0.429 | 0.340 | 0.273 |
| | Persistence | -0.166 | 0.512 | 0.390 | 0.320 | 0.225 |
| 6 h | Ensemble | **0.284** | **0.517** | 0.418 | 0.328 | 0.286 |
| | LightGBM | 0.237 | 0.494 | 0.389 | 0.310 | 0.258 |
| | LSTM | 0.281 | 0.489 | **0.440** | **0.341** | **0.297** |
| | LR | 0.215 | 0.467 | 0.388 | 0.293 | 0.245 |
| | Analog (k-NN) | 0.114 | 0.440 | 0.257 | 0.181 | 0.156 |
| | Persistence | -0.695 | 0.405 | 0.162 | 0.120 | 0.073 |

RPSS relative to the month–hour class climatology; $HSS_3$: three-category Heidke score. $CSI_{\geq II}$: RVR < 550 m (CAT II or worse); $CSI_{III}$: RVR < 350 m. LR: logistic regression; RF: random forest.

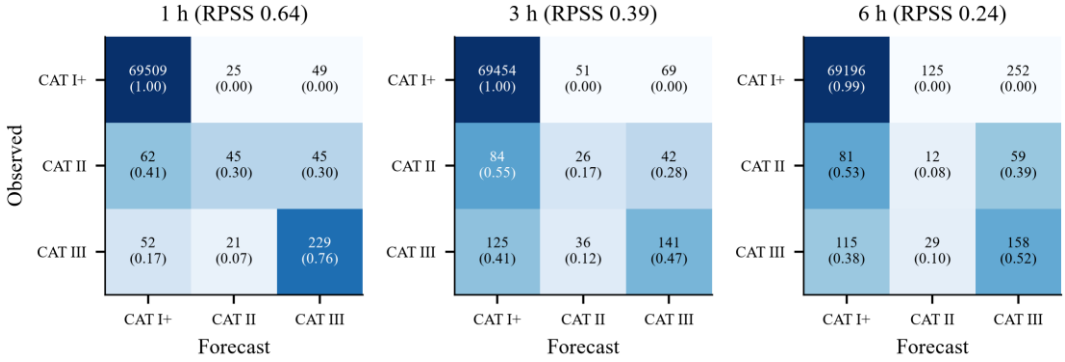



Fig. 9. Confusion matrices of the LightGBM ILS-category forecasts on the test seasons (counts and row-normalised fractions).

### *F. Class Imbalance and Calibration*

The imbalance ablation (Table XI) shows that class weighting, SMOTE and undersampling do not improve the ranking quality reliably—for LR they even lower the validation PR-AUC from 0.759 to 0.720–0.747—but they severely degrade the raw probabilities: class weighting increases the validation Brier score of LR tenfold (from 0.0026 to 0.0261). Recalibration removes most of this damage, but the simplest option, no resampling followed by calibration, is as good as or better than the alternatives; it was therefore used for all final models. The reliability curves of the calibrated forecasts follow the diagonal closely at all lead times (Fig. 10). This is not trivial: the fog frequency of the test seasons (0.64%) is only half that of the training period (1.24%), and calibrating on the most recent seasons adapted the probabilities to this change.

TABLE XI
CLASS-IMBALANCE STRATEGIES (FOG, 3 H)

| Model | Strategy | PR-$AUC_v$ | $BS_v$ raw | $BS_v$ cal. | PR-$AUC_t$ | $CSI_t$ |
|---|---|---|---|---|---|---|
| Logistic regression | none | 0.759 | 0.0026 | 0.0026 | 0.636 | 0.395 |
| | class weights | 0.720 | 0.0261 | 0.0028 | 0.594 | 0.406 |
| | SMOTE | 0.736 | 0.0101 | 0.0027 | 0.614 | 0.408 |
| | undersampling | 0.747 | 0.0056 | 0.0026 | 0.619 | 0.405 |
| LightGBM | none | 0.741 | 0.0026 | 0.0026 | 0.632 | 0.457 |
| | class weights | 0.743 | 0.0056 | 0.0025 | 0.638 | 0.462 |
| | SMOTE | 0.754 | 0.0027 | 0.0026 | 0.661 | 0.443 |
| | undersampling | 0.764 | 0.0047 | 0.0024 | 0.632 | 0.446 |

v: validation, t: test. Calibration (none/Platt/isotonic) selected by leave-one-season-out CV.

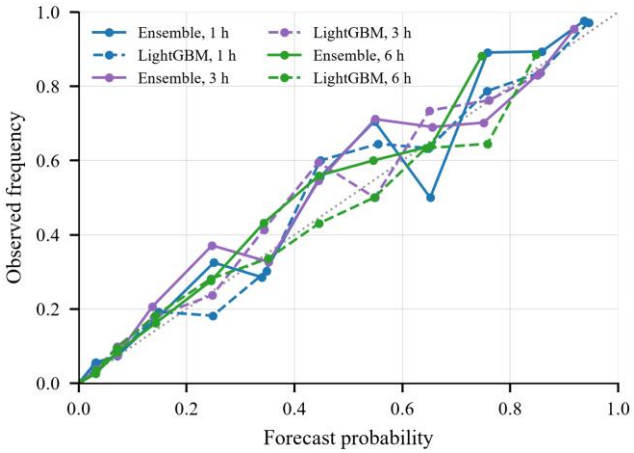



Fig. 10. Reliability diagram of the calibrated ensemble (solid) and LightGBM (dashed) fog forecasts on the test seasons; bins with fewer than 15 forecasts are omitted.

### *G. Explainability*

The SHAP analysis reveals a physically coherent picture. At 1 h the forecast is dominated by the current state—relative humidity, minimum and prevailing visibility and dew-point depression (Fig. 11)—which together carry 78% of the total attribution (Fig. 12). As the lead time increases, the share of the current state falls to 41% at 6 h, while the astronomical and calendar terms rise from 6% to 28% and boundary-layer memory from 6% to 21%.

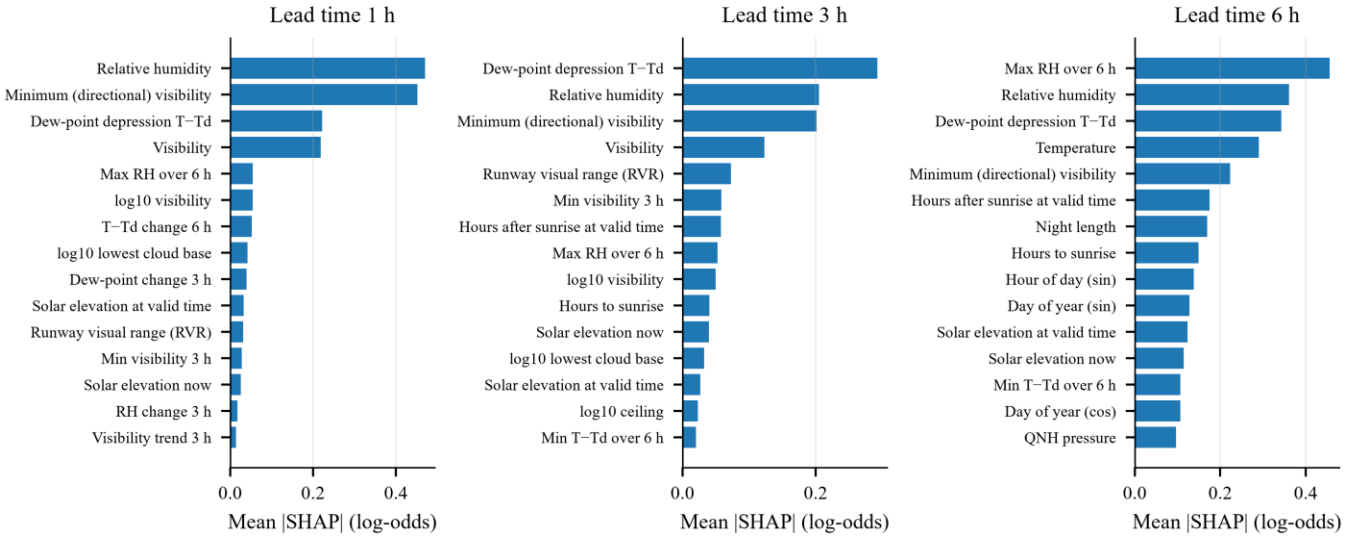



Fig. 11. Mean absolute SHAP values of the fifteen most important predictors of the LightGBM fog models at 1, 3 and 6 h.

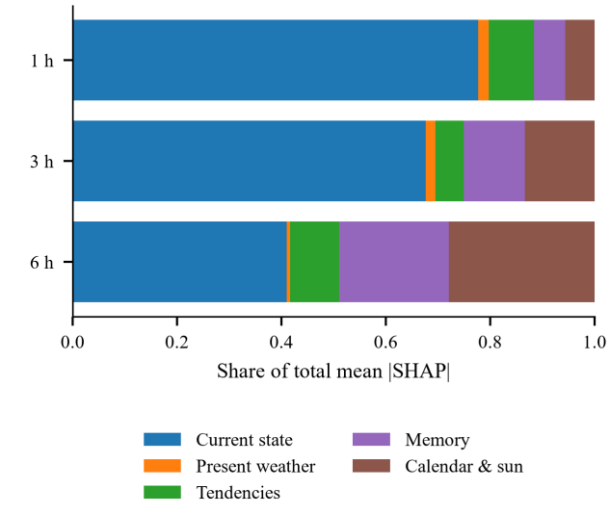



Fig. 12. Share of the total mean absolute SHAP value per predictor group and lead time.

At 6 h the 6-h maximum relative humidity, the relative humidity, the dew-point depression and the temperature lead the ranking, followed by the time since sunrise at the valid time, the night length and the time to sunrise (Fig. 13): high humidity, small dew-point depressions, low temperatures and long nights push the forecast towards fog, whereas valid times after sunrise push it away—the signature of radiation fog.

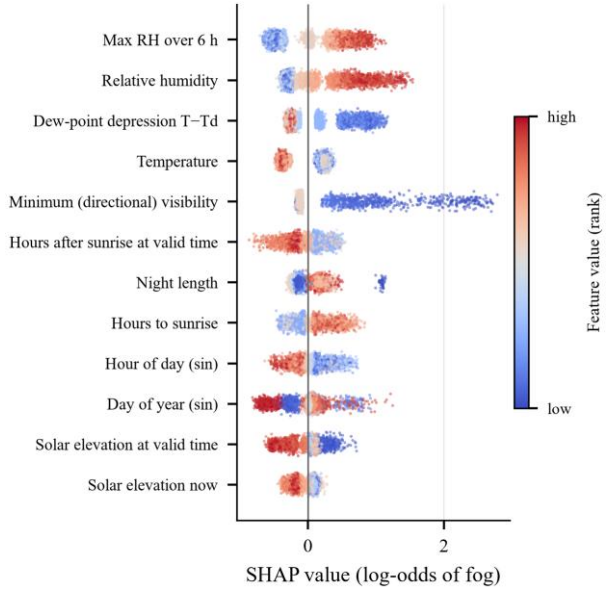



Fig. 13. SHAP values of the twelve most important predictors of the 6-h model; colour shows the percentile rank of the predictor value.

The dependence plots (Fig. 14) reproduce textbook fog

physics: dew-point depressions of 2 °C or less raise the log-odds by up to +1.4, whereas depressions of 4 °C or more lower them; calm and light winds (≤ 3 kt) favour fog, while winds of 4–10 kt suppress it through mechanical mixing; a 6-h maximum relative humidity above about 93% strongly favours fog; and once the sun is more than about 15° above the horizon at the valid time the contribution turns negative, reflecting dissipation by solar heating.

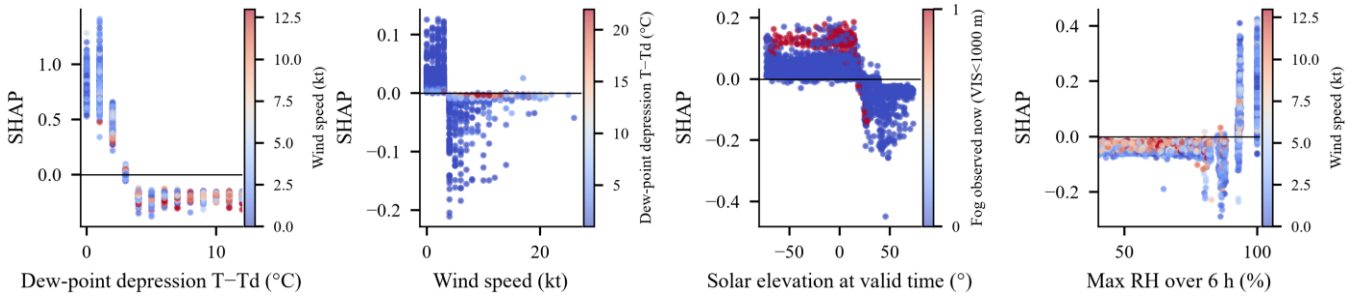



Fig. 14. SHAP dependence of the 3-h fog model on the dew-point depression, wind speed, solar elevation at the valid time and 6-h maximum relative humidity, coloured by an interacting variable.

*H. Case Study: 17–18 December 2022*

Fig. 15 shows a two-stage episode from the test period. Fog persisted until about 10 UTC on 17 December, cleared during the afternoon and formed again at 16:50 UTC, lasting until 03:50 UTC on 18 December with RVR mostly between 200 and 400 m. At 16:20 UTC, when the visibility was still 1200 m, the 3-h model issued a fog probability of 0.82 (threshold 0.33), i.e., a correct warning for the re-formation. Its SHAP explanation (Fig. 16) attributes the forecast to the falling minimum visibility (+1.74), the RVR of 650 m (+0.96), the dew-point depression of 1 °C (+0.88) and a relative humidity of 93% (+0.88)—exactly the cues an experienced forecaster would use. The 1-h forecasts track the episode closely, and the 3- and 6-h forecasts raise the probability during the evening and anticipate the persistence of the fog through the night; after sunrise all lead times correctly forecast the dissipation, with the longer lead times reacting later.

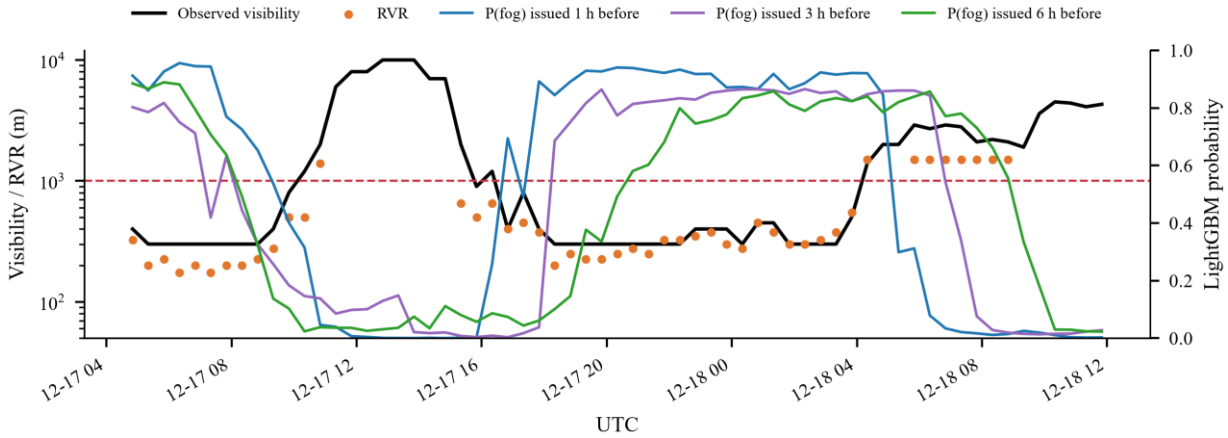



Fig. 15. Observed visibility and RVR at LTAC on 17–18 December 2022 with the LightGBM fog probabilities issued 1, 3 and 6 h before each valid time.

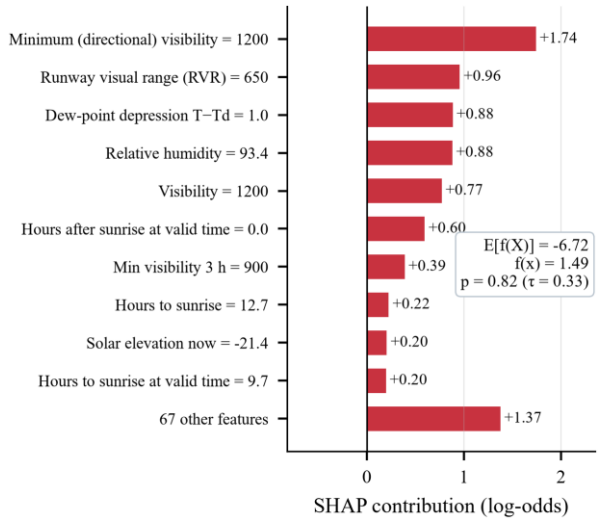



Fig. 16. SHAP explanation of the 3-h forecast issued at 16:20 UTC on 17 December 2022 (visibility 1200 m; fog formed 30 min later).

### *I. Operational Performance*

The deployed application (Fig. 17) was verified in two ways. First, a replay of the operational period after the test set (July–October 2026) through the web-application code yielded 5 727–5 737 forecasts per lead time (Table XII); no fog occurred in this warm-season period and the system issued no fog warning at any lead time, i.e., it produced no false alarms. Second, a parity check rebuilt the features at 40 random times from only the reports observed up to that time, exactly as the live system does, and reproduced the batch probabilities to machine precision (maximum difference 0.0 over 240 comparisons). The live logger started on 29 October 2026, at the beginning of the 2026/27 fog season, and its automatically verified record is updated continuously during the semester.

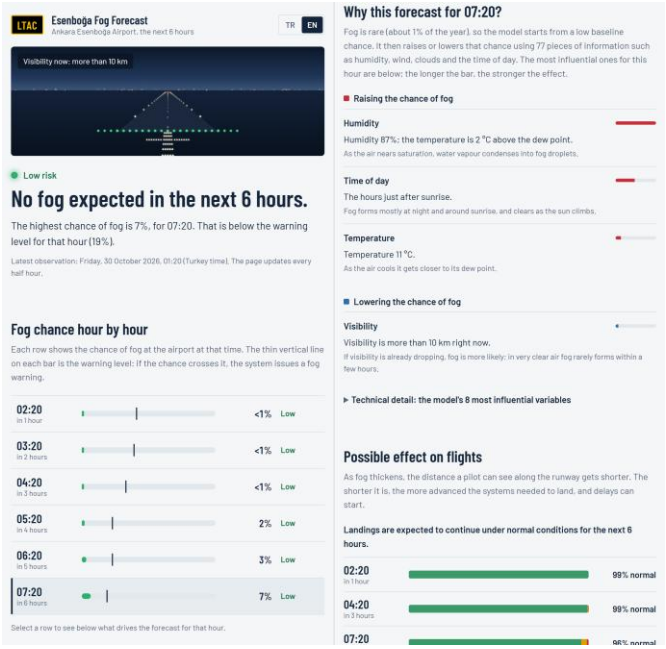



Fig. 17. The web-based early-warning application. Left: plain-language status over a runway scene drawn from the observed visibility, and the fog probabilities for the next 6 h against their warning thresholds. Right: the driving factors of the selected hour (SHAP values summed into plain-language groups) and the ILS-category outlook.

TABLE XII
OPERATIONAL VERIFICATION AFTER THE TEST PERIOD

| Period | Lead | n | Fog | Warnings | POD | FAR | BS |
|---|---|---|---|---|---|---|---|
| 07/2026–10/2026 replay | 1 h | 5 737 | 0 | 0 | – | – | 0.0000 |
| 07/2026–10/2026 replay | 2 h | 5 735 | 0 | 0 | – | – | 0.0000 |
| 07/2026–10/2026 replay | 3 h | 5 734 | 0 | 0 | – | – | 0.0000 |
| 07/2026–10/2026 replay | 4 h | 5 731 | 0 | 0 | – | – | 0.0000 |
| 07/2026–10/2026 replay | 5 h | 5 729 | 0 | 0 | – | – | 0.0000 |
| 07/2026–10/2026 replay | 6 h | 5 727 | 0 | 0 | – | – | 0.0000 |

Replay: archived METARs processed with the web-application code; live: forecasts logged by the running application and verified automatically; catch-up: forecasts for issue times skipped by the scheduler, computed in the next run from the reports observed up to that time (later corrections included).

## VI. DISCUSSION

The results show that a purely observation-based nowcast can add substantial skill to both persistence and the official TREND at Esenboğa, and that this gain is much larger than the differences among modern ML techniques. Four aspects deserve emphasis. First, the best family depends on the lead time: at 1 h the tree ensembles and the EBM, which exploit sharp thresholds in the current state, are best, whereas at 3–6 h the recurrent networks extract more from the 12-h history. This agrees with the observation that trees remain hard to beat on tabular predictors [23], while sequence models pay off once the evolution of the boundary layer matters. Second, an ensemble of diverse models is the most consistent choice across lead times and scores, at the price of opacity and of training thirteen models; for an operational system with limited resources LightGBM, second overall and the fastest to train, is a sound single model, which is why it powers the web application. Third, the EBM shows that a transparent model can be practically as accurate as the best one at 1 h, and the SHAP analysis confirms that the models learn the physics of radiation fog—saturation, weak mixing and the diurnal radiative cycle—rather than spurious correlations, which supports forecaster trust [26] and agrees with the dominance of radiation fog at LTAC [4]. Fourth, the comparison with the TREND shows that the official 2-h forecast is essentially persistence, and the simple rule of thumb over-forecasts fog, so even modest data-driven guidance can improve the very-short-range service; the economic-value analysis shows that this holds for a wide range of users.

The study has limitations. It uses point observations only; fog onsets, which depend on boundary-layer processes and advection that are not visible in a single METAR, remain hard to predict, and numerical weather prediction, satellite or ceilometer data would likely improve them [11], [19]. The historical TAF archive could not be obtained for the test seasons, so the TAF comparison is limited to a short, fog-free window; it will be completed with the live verification.

Because only one airport is considered, transferability is not assessed [20]. With four test seasons the differences among the leading techniques are mostly within the sampling uncertainty, so their ranking may change with more data; tabular foundation models [24], which cannot use 250 000 training cases without subsampling, were not included. Finally, the decreasing frequency of fog over the 21 years implies non-stationarity; the season-aligned validation and calibration on recent seasons mitigated it, but periodic retraining will be necessary.

## VII. Conclusion

This paper presented the first machine-learning nowcasting system for fog and low visibility at Ankara Esenboğa Airport and a benchmark of thirteen techniques from six families. Built on 21.8 years of quality-controlled METAR/SPECI reports and verified on four independent fog seasons, all techniques significantly outperformed persistence; a greedy ensemble ranked first overall, tree ensembles and the glass-box EBM led at 1 h and recurrent networks at 3–6 h, and LightGBM improved on the official METAR TREND by 0.106 in CSI at 1 h. Calibrated probabilities, ILS-category forecasts, economic-value curves and SHAP explanations make the forecasts operationally interpretable, and the open-source web application delivers them in real time with continuous self-verification. Future work will add numerical-model and satellite predictors to improve onset detection, extend the TAF comparison over full fog seasons, test tabular foundation models and transfer the approach to other Turkish airports.

## Appendix

The hyper-parameters selected on the validation seasons are listed in Table XIII. The complete data set, the decoder, training and evaluation scripts, the trained models, the figures in English and Turkish and the web application are available at https://github.com/AtaSakik/esenboga-fog-nowcasting; a single script reproduces every number in this paper.

TABLE XIII
SELECTED HYPER-PARAMETERS

| Model | Lead | Setting |
|---|---|---|
| CatBoost | 1 h | η=0.078, depth=7, l2=9.84, random strength=0.21, bagging T=0.06 |
| CatBoost | 3 h | η=0.058, depth=8, l2=3.84, random strength=0.07, bagging T=0.23 |
| CatBoost | 6 h | η=0.092, depth=7, l2=3.11, random strength=1.66, bagging T=0.96 |
| LightGBM | 1 h | η=0.026, leaves=17, min_child=42, ff=0.76, bf=0.71, $\lambda_2$=0.0383 |
| LightGBM | 2 h | η=0.027, leaves=119, min_child=203, ff=0.96, bf=0.95, $\lambda_2$=0.475 |
| LightGBM | 3 h | η=0.025, leaves=27, min_child=59, ff=0.67, bf=0.89, $\lambda_2$=0.00783 |
| LightGBM | 4 h | η=0.052, leaves=16, min_child=122, ff=0.50, bf=0.53, $\lambda_2$=17.7 |
| LightGBM | 5 h | η=0.052, leaves=16, min_child=122, ff=0.50, bf=0.53, $\lambda_2$=17.7 |
| LightGBM | 6 h | η=0.053, leaves=15, min_child=316, ff=0.69, bf=0.69, $\lambda_2$=14 |
| XGBoost | 1 h | η=0.032, depth=5, mcw=2.0, subsample=0.70, colsample=0.74, λ=0.447 |
| XGBoost | 3 h | η=0.035, depth=9, mcw=14.8, subsample=1.00, colsample=0.70, λ=0.574 |
| XGBoost | 6 h | η=0.084, depth=6, mcw=8.1, subsample=0.60, colsample=0.45, λ=27.7 |
| Analog (k-NN) | 1/3/6 | 1: k=200; 3: k=200; 6: k=200 |
| EBM | 1/3/6 | 1: interactions=20, η=0.03; 3: interactions=10, η=0.015; 6: interactions=20, η=0.03 |
| MLP | 1/3/6 | 1: 2×128, dropout=0.1, lr=0.001; 3: 4×256, dropout=0.3, lr=0.0003; 6: 4×256, dropout=0.3, lr=0.0003 |
| GRU | 1/3/6 | hidden=32, layers=2, dropout=0.2, lr=0.001, wd=0.0001, seq=24×30 min |
| LSTM | 1/3/6 | hidden=64, layers=2, dropout=0.2, lr=0.001, wd=0.0001, seq=24×30 min |
| TCN | 1/3/6 | hidden=32, layers=2, dropout=0.2, lr=0.001, wd=0.0001, seq=24×30 min |
| Transformer | 1/3/6 | hidden=32, layers=2, dropout=0.2, lr=0.001, wd=0.0001, seq=24×30 min |

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